

Question Option B				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(a)		(i)	cartilage (1) Chondrocytes (1)	2			2		
			(ii)	Compact (bone) and Calcium phosphate/hydroxyapatite (1)	1			1		
	(b)		(i)	Osteoblasts – build up bone and Osteoclasts – break down bone (1)	1			1		
			(ii)	Oestrogen would decrease osteoclast activity/bring osteoclast activity to normal level/reduce loss calcium from bones(1) Less bone broken down (1)			2	2		
			(iii)	Vitamin D supplements/Calcium supplements (1) increases calcium absorption (in the gut)/increase bone formation (1)		2		2		
			(iv)	10 yr old girls have same BMD as 10 yr olds with TS (1) Difference between normal and TS not obvious til 12+ years (1)			2	2		1
			(v)	Compares the result with the mean/Shows how far the value is from the mean		1		1		1
			(vi)	Bone realignment/immobilisation (in a cast or splint)	1			1		
			(vii)	Less calcium ions bind to troponin so no shape change (1) (No shape change results in) less tropomyosin being moved (1) Exposing less myosin binding sites (on the actin) (1) Resulting in less force exerted (1)		3	1	4		
	(c)		(i)	Third (order lever)	1			1		
			(ii)	333.2/333 =2 marks 39.2 x (34/4) (1)		2		2	2	
			(iii)	Age of the patients/other health issues/general fitness		1		1		1
				Question 7 Option B total	6	9	5	20	2	3